



## PH 621: Computational Methods for Physicists

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|---------------|--------------------------------------------------|
| Course Code   | PH 621                                           |
| Course Name   | Computational Methods for Physicists             |
| L-T-P-C       | 2-0-4-4                                          |
| Prerequisites | Faculty consent                                  |
| Intended for  | I-Ph.D., M.Sc., B.Tech 3rd and 4th Year.         |
| Distribution  | Core for I-Ph.D.; Elective for the rest          |
| Approval      | 10th Senate (Jun 2016)                           |
| Revised       | 27th BoA (Nov 2018) - credits changed to 2-0-4-4 |

### Course Contents

- **Introduction:** Basic introduction to operating system fundamentals [4 Lectures]
- **Introduction to C:** Program Organization and Control Structures loops, arrays, and function, Error, Accuracy, and Stability. [8 Lectures]
- **Interpolation and Extrapolation - Curve Fitting:** Polynomial Interpolation and Extrapolation Cubic Spline Interpolation Fitting Data to a Straight Line, examples from experimental data fitting [8 Lectures]
- **Integration and differentiation:** Numerical Derivatives Romberg Integration Gaussian Quadratures and Orthogonal Polynomials [8 Lectures]
- **Root Finding:** Newton-Raphson Method Using Derivative - Roots of a Polynomial [8 Lectures]
- **Ordinary Differential Equations:** Runge-Kutta Method, Adaptive Stepsize Control for Runge-Kutta, Examples from electrodynamics and quantum mechanics [8 Lectures]
- **Matrices and algebraic equations:** Gauss-Jordan Elimination Gaussian Elimination with Backsubstitution, LU Decomposition [8 Lectures]
- **Module VIII:** Concept of simulation, random number generator [2 Lectures]

### Text Books

1. B W Kernighan and D M Richie, *The C Programming Language*, PHI Learning, 2011.
2. S D Conte and C de Boor, *Elementary numerical analysis: algorithmic approach*, McGraw-Hill International, 1980.

### References

1. V. Rajaraman, *Computer Programming in C*, PHI Learning Pvt. Ltd, 2011.
2. Germund Dalquist and Ake Bjork, *Numerical Methods*, Dover Publications, 1974.
3. William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery, *Numerical Recipes*, Cambridge University Press, 1992.